

Verahexal® 120 SR

Sustained-Released Tablet

CONTENT

Each sustained-released tablet contains 120mg Verapamil Hydrochloride.

DESCRIPTION

Verahexal® 120 SR tablets are beige to ochre colour with no dots or spots of different intensities and colours. The tablet has a smooth surface with a round, slightly biconvex shape, scored on one side, with diameter of 11.0-11.2mm and thickness of 4.4-4.8mm.

INDICATION

Verahexal® 120 SR is indicated for:

- The treatment of mild to moderate hypertension.
- The treatment and prophylaxis of angina pectoris.
- Secondary prevention of re-infarction after an acute myocardial infarction in patients without heart failure, and not receiving diuretics (apart from low-dose diuretics when used for indications other than heart failure), and where beta-blockers are not appropriate. Treatment is to be started at least one week after an acute myocardial infarction.

RECOMMENDED DOSAGE

Adults

Hypertension: The usual dose is Verapamil Hydrochloride SR 240mg daily. For patients new to verapamil therapy the doctor should consider halving the dose to Verapamil Hydrochloride SR 120mg daily. For most patients respond to 240mg daily given as a single dose. If control is not achieved within at least one week the dosage may be increased to maximum of 480mg daily (two **Verahexal® 120 SR** tablets taken in the morning and two **Verahexal® 120 SR** tablets in the evening at an interval of about 12 hours). Other anti-hypertensive agents, in particular diuretics, may be used in combination with Verapamil Hydrochloride SR 240mg tablets to achieve further reduction in blood pressure. **Verahexal® 120 SR** tablets may be used for dose titration purposes.

Angina pectoris: The usual dose is Verapamil Hydrochloride SR 240mg twice daily. A small number of patients respond to a lower dose and where indicated, adjustment down to Verapamil Hydrochloride SR 240mg tablet daily could be made. **Verahexal® 120 SR** tablets may be used for dose titration purposes.

Secondary prevention of re-infarction after an acute myocardial infarction in patients without heart failure, and not receiving diuretics (apart from low-dose diuretics when used for indications other than heart failure), and where beta-blockers are not appropriate: Treatment is to be started at least one week after an acute myocardial infarction. 360mg/day in divided doses, to be taken either as 120mg three times daily or as 240mg to be taken in the morning and 120mg to be taken in the evening, on a daily basis.

Elderly patients

The adult dose is recommended, except in patients with impaired liver or renal function or cardiac conduction problems where a reduced dosage may be necessary (see **WARNINGS AND PRECAUTIONS** section).

Verapamil therapy should be discontinued gradually after long-term treatment.

Method of administration

For oral administration.

Tablets to be taken without sucking and chewing together with sufficient liquid (e.g., water) DO NOT take with grapefruit juice.

WARNINGS AND PRECAUTIONS

Acute Myocardial infarction: Use with caution in acute myocardial infarction with complications (bradycardia, hypotension, left ventricular dysfunction).

Conduction disturbance/ AV block degree I/ bradycardia/asystole: Verapamil hydrochloride affects the AV and sinus node and prolongs AV conduction time. Use with caution as it may produce second- or third-degree AV block, or unifascicular, bifascicular or trifascicular bundle branch block requires discontinuation of verapamil hydrochloride and institution of appropriate therapy, if needed.

Verapamil hydrochloride affects the AV and sinus nodes and rarely may produce second- or third-degree AV block, bradycardia, or in extreme cases asystole. This is more likely to occur in patients with a sick sinus syndrome, which is more common in older patients.

Asystole in patients other than those with sick sinus syndrome is usually of short duration (few seconds or less), with spontaneous return to AV nodal or normal sinus rhythm. If this does not occur promptly, appropriate treatment should be initiated immediately.

Antiarrhythmics, beta-blockers and inhalation anaesthetics:

Antiarrhythmics (e.g., flecainide, disopyramide), beta-receptor blockers (e.g., metoprolol, propranolol) and inhalation anaesthetics may cause mutual potentiation of cardiovascular effects (higher-grade AV block, higher grade lowering of heart rate, induction of heart failure and potentiated hypotension).

Asymptomatic bradycardia (36 beats/minute) with a wandering atrial pacemaker has been observed in a patient receiving concomitant timolol (a beta-adrenergic blocker) eye drops and oral verapamil hydrochloride.

Digoxin: If verapamil is administered concomitantly with digoxin, digoxin dose is to be reduced (See **DRUG INTERACTIONS** section).

Cardiac insufficiency: Cardiac insufficiency patients with ejection fraction higher than 35% should be compensated before starting verapamil treatment and should be adequately treated throughout.

Neuromuscular transmission disorders: Used with caution in the presence of diseases in which neuromuscular transmission is affected (myasthenia gravis, Lambert-Eaton syndrome, advanced Duchenne muscular dystrophy).

HMG-CoA reductase inhibitor: Used in caution for patients taking statins.

Hypotension: In hypotension (less than 90mmHg systolic blood pressure) extra careful monitoring is required.

Special populations

Impaired renal function: Although comparison studies have reliably shown that impaired renal function in patients with end-stage renal failure has no effect on the pharmacokinetics of verapamil, isolated case reports have suggested that verapamil should only be used with caution and with careful monitoring (ECG, blood pressure) in patients with impaired renal function.

Verapamil cannot be removed via haemodialysis.

Impaired hepatic function: Used with caution in patients with severely impaired hepatic function.

Use in Paediatric: The safety and efficacy of verapamil prolonged release tablets in children and adolescents have not been established. No data are available.

CONTRAINDICATIONS

Verapamil should not be used in cases of

- Hypersensitivity (allergy) to the active substance verapamil hydrochloride or any of the excipients mentioned.
- Cardiovascular shock
- Pronounced conduction disturbances (such as SA or AV block degrees II and III; except in patients with an artificial pacemaker)
- Sick sinus syndrome (except in patients with an artificial pacemaker)
- Cardiac insufficiency with reduced ejection fraction of less than 35% and/or pulmonary wedge pressure above 20mmHg (unless secondary to a supraventricular tachycardia amenable to verapamil therapy)
- Atrial fibrillation/flutter in the presence of an accessory bypass tract (e.g., Wolff-Parkinson-White syndrome or Lown-Ganong-Levine syndrome). These patients are at risk to develop ventricular tachyarrhythmia including ventricular fibrillation if verapamil hydrochloride is administered.
- Existing treatment with Ivabradine (for heart conditions)
- Beta-blocker must not be used concurrently intravenously in patients treated with verapamil (exception intensive care medicine)

DRUG INTERACTIONS

Concomitant medicinal product	Potential effect on verapamil or concomitant medicinal product	Comments
Alpha Blockers		
Prazosin	↑ prazosin C_{max} (~40%) with no effect on half-life	Additive hypotensive effect
Terazosin	↑ Terazosin AUC (~24%) and C_{max} (~25%)	
Antiarrhythmics		
Flecainide	Minimal effect on flecainide plasma clearance (<~10%); no effect on verapamil plasma clearance	Further information see WARNINGS AND PRECAUTIONS section
Quinidine	↓ oral quinidine clearance (~35%)	
Amiodarone	Increase of amiodarone plasma levels	
Antiasthmatics		
Theophylline	↓ oral and systemic clearance by ~20%	Reduction of clearance was lessened in smokers (~11%)
Anticonvulsants		
Carbamazepine	↑ carbamazepine AUC (~46%) in refractory partial epilepsy patients	Increased carbamazepine levels. This may produce carbamazepine side effects such as diplopia, headache, ataxia or dizziness
	Decrease of plasma verapamil hydrochloride level	
Phenytoin	↓ verapamil plasma concentrations	
Antidepressants		
Imipramine	↑ Imipramine AUC (~15%)	No effect on level of active metabolite, desipramine
	Increase of verapamil hydrochloride plasma level	
Antidiabetics		
Glibenclamide	↑ glyburide C_{max} (~28%), AUC (~26%)	
	Increase verapamil plasma level	
Antigout agents		
Colchicine	↑ colchicine AUC (~2.0 fold) and C_{max} (~1.3 fold)	Reduction of colchicine dose (concomitant administration of colchicine and verapamil is not recommended)
Anti-infectives		
Clarithromycin Erythromycin Telithromycin	Possible ↑ in verapamil levels	
Rifampicin	↓ verapamil AUC (~97%), C_{max} (~94%), oral bioavailability (~92%) with oral verapamil administration	Blood pressure lowering effect may be reduced
Anti-neoplastic		
Doxorubicin	↑ doxorubicin AUC (104%) and C_{max} (61%) with oral verapamil administration	In patients with small cell lung cancer
Anti-fungal		
Clotrimazole Ketoconazole Itraconazole	Increase of verapamil hydrochloride plasma level	
Barbiturates		
Phenobarbital	↑ oral verapamil clearance (~5 fold)	
Benzodiazepines and other anxiolytics		
Buspirone	↑ buspirone AUC, C_{max} by ~3.4 fold	
	Increase of verapamil plasma level	
Midazolam	↑ midazolam AUC (~3 fold) and C_{max} (~2 fold)	
	Increase of verapamil hydrochloride plasma level	
Beta blockers		
Metoprolol	↑ metoprolol AUC (~32.5%) and C_{max} (~14%) in angina pectoris patients	See WARNINGS AND PRECAUTIONS section
	Increase of verapamil plasma levels	

Propranolol	↑ propranolol AUC (~65%) and C _{max} (~94%) in angina pectoris patients	
	Increase of verapamil plasma level	
Cardiac glycoside		
Digitoxin	↓ digitoxin total body clearance (~27%) and extra renal clearance (~29%)	
Digoxin	Healthy subjects: ↑ C _{max} (~44%) ↑ digoxin C _{12h} (~53%) ↑ C _{ss} (~44%) ↑AUC (~50%)	Reduction of digoxin dose.
Immuno-suppressive		
Ciclosporin	↑ ciclosporin AUC, C _{ss} , C _{max} by ~45%	
Everolimus	Everolimus: ↑ AUC (~3.5 fold) and ↑ C _{max} (~2-3 fold) Verapamil: ↑ C _{trough} (~2.3 fold)	Concentration determinations and dose adjustments of everolimus may be necessary
Sirolimus	Sirolimus ↑ AUC (~2.2 fold), S- Verapamil AUC: ↑ (~1.5 fold)	Concentration determinations and dose adjustments of Sirolimus may be necessary
Tacrolimus	Possible ↑ tacrolimus levels	
Lipid lowering agents (HMG CoA reductase inhibitors)		
Atorvastatin	Possible ↑ atorvastatin levels increase verapamil AUC (~43%)	Additional information see below
Lovastatin	Possible ↑ lovastatin levels increase verapamil AUC (~63%) and C _{max} (~32%)	
Simvastatin	↑ simvastatin AUC (~2.6 fold), C _{max} (~4.6 fold)	
H ₂ antagonist receptor		
Cimetidine	↑ AUC of R (~25%) and S (~40%) verapamil with corresponding ↓ in R- and S-verapamil clearance	Cimetidine reduces verapamil clearance following intravenous verapamil administration
Serotonin receptor antagonists		
Almotriptan	↑ almotriptan AUC (~20%) ↑ C _{max} (~24%)	
	Increase of verapamil hydrochloride plasma levels	
Uricosurics		
Sulfinpyrazone	↑ verapamil oral clearance (~3 fold) ↓ bioavailability (~60%)	Blood pressure lowering effect may be reduced.
Other cardiac treatments agents		
Ivabradine	Concomitant use with ivabradine is contraindicated due to the additional heart rate lowering effect of verapamil to ivabradine	
Others		
Grapefruit juice	↑ R- (~49%) and S- (~37%) verapamil AUC ↑ R- (~75%) and S- (~51%) verapamil C _{max}	Elimination half-life and renal clearance not affected. Food and beverage containing grapefruit should be avoided during use of verapamil
St. John's Wort	↓ R- (~78%) and S- (~80%) verapamil AUC with corresponding reduction with C _{max}	

Other drug interactions and additional drug interaction information

HIV antiviral agents

Due to the metabolic inhibitory potential of some of the HIV antiviral agents, such as ritonavir, plasma concentrations of verapamil may increase. Caution should be used and the dose of verapamil may be decreased.

Correspondingly, verapamil may induce increased plasma level of these medicinal agents by influencing degradation.

Lithium

Increased sensitivity to the effects of lithium (neurotoxicity) has been reported during concomitant verapamil hydrochloride-lithium therapy with either no change or an increase in serum lithium levels.

The addition of verapamil hydrochloride, however, has also

resulted in the lowering of the serum lithium levels in patients receiving chronic oral lithium. Patients receiving both medicinal products should therefore be monitored carefully.

Muscle relaxants

Clinical data and animal studies suggest that verapamil hydrochloride may potentiate the activity of muscle relaxants (curare-like and depolarizing). It may be necessary to decrease the dose of verapamil and/or the dose of the neuromuscular blocking agent when the medicinal products are used concomitantly.

Acetylsalicylic acid

Increased haemorrhagic diathesis.

Dabigatran

When oral verapamil was co-administered with dabigatran etexilate (150mg), a P-gp substrate, the C_{max} and AUC of dabigatran were increased but magnitude of this change differs depending on time between administration and the formulation of verapamil.

Co-administration of verapamil 240mg extended release at the same time as dabigatran etexilate resulted in increased dabigatran exposure (increase of C_{max} by about 90% and AUC by about 70%).

Close clinical surveillance is recommended when verapamil is combined with dabigatran etexilate and particularly in the occurrence of bleeding, notably in patients having a mild to moderate renal impairment.

Other direct oral anticoagulants (DOACs)

Increased absorption of DOACs since they are P-gp substrates and, if applicable, also reduced elimination of DOACs which are metabolized by CYP3A4, may increase the systemic bioavailability of DOACs. Some data suggest a possible increase of the risk of bleeding, especially in patients with further risk factors.

(see DOAC label for further information)

Ethanol (alcohol)

Delayed ethanol elimination and elevation of ethanol plasma levels, resulting in enhanced effect of alcohol by verapamil.

HMG Co-A reductase inhibitors (statins)

In patients taking verapamil, treatment with HMG-CoA reductase inhibitors (e.g., simvastatin, atorvastatin or lovastatin) should be started at the lowest possible dose and titrated upwards. If verapamil treatment is to be added to patients already taking an HMG CoA reductase inhibitor (e.g., simvastatin, atorvastatin or lovastatin), a reduction in the statin dose should be considered, whereby the dose should be re-titrated based on serum cholesterol concentrations.

If verapamil and simvastatin are used concurrently at higher doses, the risk of myopathy/rhabdomyolysis is increased. The dose of simvastatin should be adjusted accordingly (see product information of the manufacturer, and **WARNINGS AND PRECAUTIONS** section)

Fluvastatin, pravastatin and rosuvastatin are not metabolized by CYP3A4 and are less likely to interact with verapamil.

Antihypertensive, diuretics, vasodilators

Increase of the antihypertensive effect with risk of accelerated fall in blood pressure

Antiarrhythmics (e.g., flecainide, disopyramide), beta-adrenergic blocking agents (e.g., metoprolol, propranolol), inhalation anaesthetics

Reciprocal increase in cardiovascular effects (higher-degree AV block, higher-degree decrease in the heart rate, occurrence of cardiac insufficiency, potentiated fall in blood pressure).

Beta-blockers must not be used concurrently intravenously in patients treated with verapamil (exception intensive care medicine).

Metformin

Co-administration of verapamil with metformin may reduce the efficacy of metformin.

PREGNANCY AND LACTATION

Pregnancy

Verapamil hydrochloride has placental patency. The plasma concentration in umbilical venous blood is 20-92% of the maternal plasma concentration. No sufficient experience is available regarding use of verapamil hydrochloride during pregnancy. However, data about a

limited number of pregnant women treated orally do not suggest teratogenic effects of verapamil hydrochloride.

Verapamil hydrochloride should therefore it should not be taken in the first or second trimester of pregnancy. It may be taken in the third trimester only in case of compelling indication while considering the risk for mother and child.

Lactation

Verapamil is excreted in breast milk (milk concentration is approximately 23% of the maternal plasma concentration). Limited human data after oral ingestion have shown that the infant absorbs only a small amount of the active substance (0.1 to 1% of the maternal dose) so taking verapamil could therefore be compatible with breast feeding.

A risk for the newborn/infant cannot be excluded.

Due to risk of severe adverse reactions in infants, verapamil should not be administered during lactation unless absolute necessary for welfare of the mother.

Evidence exists that verapamil can cause hyperprolactinaemia and galactorrhoea in isolated cases.

SIDE EFFECTS

The most commonly reported ADRs were headache, dizziness and drowsiness, gastrointestinal disorders (nausea, constipation and abdominal disorders), further bradycardia, tachycardia, palpitations, hypotension, flushing, oedema peripheral, as well as fatigue.

Immune system disorders

Not known: Hypersensitivity

Metabolism and nutrition disorders

Uncommon: Reduced glucose tolerance

Not known: Hyperkalaemia

Psychiatric disorders

Common: Nervousness

Rare: Somnolence

Nervous system disorders

Common: Dizziness or drowsiness, headache, neuropathy

Rare: Paraesthesia, tremor

Not known: Extrapyrimal symptoms, paralysis (tetraparesis), convulsions

Ear and labyrinth disorders

Rare: Tinnitus

Not known: Vertigo

Cardiac disorders

Common: Bradycardia, development of cardiac insufficiency or exacerbation of pre-existing cardiac insufficiency, excessive fall in blood pressure and/or orthostatic dysregulation

Uncommon: Palpitations, tachycardia

Not known: AV block (degree I, II, III), cardiac insufficiency, sinus arrest, sinus bradycardia, asystole

Vascular disorders

Common: Flush, hypotension

Respiratory, thoracic and mediastinal disorders

Not known: Bronchospasm, dyspnoea

Gastrointestinal disorders

Common: Constipation, nausea

Uncommon: Abdominal pain

Rare: Vomiting

Not known: Abdominal disorders, gingival hyperplasia, ileus

Hepatobiliary disorders

Uncommon: Probably allergy-induced hepatitis with reversible increase in liver-specific enzymes

Skin and subcutaneous tissue disorders

Common: Erythromelalgia

Rare: Hyperhidrosis

Very rare: Photodermatitis

Not known: Angioedema, Stevens-Johnson syndrome, erythema multiforme, alopecia, itching, pruritus, purpura, maculopapular exanthema, urticaria

Musculoskeletal and connective tissue disorders

Very rare: Exacerbation of myasthenia gravis, of Lambert-Eaton syndrome and of advanced Duchenne muscular dystrophy

Not known: Arthralgia, myasthenia, myalgia

Renal and urinary disorders

Not known: Renal insufficiency

Reproductive system and breast disorders

Not known: Erectile dysfunction, galactorrhoea, gynaecomastia

General disorders and administration site conditions

Common: Peripheral oedema

Uncommon: Fatigue

Investigations

Not known: Prolactin in blood increase

There has been a single post-marketing report of paralysis (tetraparesis) associated with the combined use of verapamil and colchicine. This may have been caused by colchicine crossing the blood-brain barrier due to CYP3A4 and P-gp inhibition by verapamil. See **DRUG INTERACTIONS** section.

Note

In patients with pacemaker, an increase in the pacing and sensing threshold cannot be excluded on verapamil.

In patients with a history of cardiovascular disease, e.g., severe cardiomyopathy, congestive heart failure or recent myocardial infarction, the risk of severe side effect was increased with concomitant administration of intravenous beta-blocker or disopyramide with intravenous verapamil as both substance classes exert a cardiodepressive effect.

SYMPTOMS AND TREATMENT OF OVERDOSE

Symptoms

The symptoms of intoxication after intoxications with verapamil depends on the quantity supplied, time of detoxification measures and myocardial contractility (age-dependent)

The following symptoms observed in case of severe intoxication:

Considerable fall in blood pressure, cardiac insufficiency, acute respiratory distress syndrome, bradycardia or tachycardia arrhythmias (e.g., junctional rhythm with AV dissociation and higher degree AV block) which can lead to cardiovascular shock and cardiac arrest.

Depression of consciousness up to coma, hyperglycaemia, hypokalaemia, metabolic acidosis, hypoxia, cardiogenic shock with pulmonary oedema, impaired renal function and convulsions, deaths have uncommonly been reported.

Treatment

Elimination of the noxae and restoration of stable cardiovascular conditions have the therapeutic priority.

The therapeutic measures depend on the time and mode of administration as well as on the type and severity of the intoxication symptoms.

In case of intoxications with higher amounts of retard preparations it has to be taken into account that the substance may still be released and absorbed in the intestine for more than 48 hours after intake.

Gastric lavage is to be recommended after oral intoxication with verapamil hydrochloride, even later than 12 hours after intake if no gastrointestinal motility (bowel sounds) can be detected. In case of suspected intoxication with retard preparations, extensive elimination measures are indicated, such as induced vomiting, removal of the contents of stomach and small intestine via suction while monitoring endoscopically, intestinal lavage, laxation, high enema.

Haemodialysis is not useful, since verapamil hydrochloride is not dialysable; however, haemofiltration and possibly plasmapheresis are recommended (high plasma protein binding of calcium antagonists).

Usual intensive care resuscitative measures, such as extrathoracic cardiac massage, ventilation, defibrillation or pace-maker therapy.

Specific measures

Removal of cardiodepressive effects, of hypotension and bradycardia.

Bradycardiac arrhythmias are to be treated symptomatically with atropine and/or beta-sympathomimetics (isoprenaline, orciprenaline); alarming cases of bradycardiac arrhythmias require temporary pace-maker therapy.

Asystole should be handled by the usual measures including beta adrenergic stimulation (e.g., isoprenaline).

Calcium is a specific antidote, e.g., 10-20 ml of a 10% calcium gluconate solution administered intravenously

(2.25 to 4.5 mmol), if necessary repeated or as continuous instillation (e.g., 5 mmol/hour).

Hypotension resulting from cardiogenic shock and arterial vasodilation is to be treated with dopamine (up to 25 µg per kg bodyweight per minute), dobutamine (up to 15 µg per kg bodyweight per minute), epinephrine or norepinephrine, respectively. The dose of these medicinal agents is orientated solely to the achieved effect. The serum calcium level should be maintained at a highly normal to slightly elevated level. Due to arterial vasodilation, liquid is additionally substituted in the early phase (Ringer's solution or sodium chloride solution).

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Treatment with verapamil requires regular medical monitoring. Due to inter-individually different reactions, reactivity may be impaired to such an extent that the ability to drive, to operate machinery or to work in an unsafe posture is impaired. This applies to a higher degree when starting therapy, increasing the dose and changing the preparation as well as in conjunction with alcohol.

Verapamil may increase the blood levels of alcohol and slow its elimination. Therefore, effects of alcohol may be exaggerated.

PHARMACODYNAMICS

Pharmacotherapeutic group: Calcium channel blockers, selective calcium channel blockers with direct cardiac effects, phenylalkylamine derivatives

ATC code: C08DA01

Verapamil hydrochloride belongs to the group of calcium antagonists. These substances have an inhibitory effect on the calcium influx via muscle cell membranes.

Verapamil hydrochloride also acts as a calcium antagonist on smooth musculature, particularly in the area of vessels and gastrointestinal tract. The effect on the smooth vascular muscles is manifested as vasodilation.

As a calcium antagonist, verapamil hydrochloride also exerts a marked effect on the myocardium. The effect on the AV node is seen in the form of prolonged conduction time. A negative-inotropic effect can occur on the working myocardium.

In humans, verapamil hydrochloride causes a decrease in total peripheral resistance due to vasodilation; no increase in cardiac output by reflex action occurs. The result is a corresponding fall in blood pressure.

PHARMACOKINETICS

Verapamil hydrochloride is a racemic mixture consisting of equal portions of the R-enantiomer and the S-enantiomer. Verapamil is extensively metabolised. Norverapamil is one of 12 metabolites identified in urine, has 10 to 20% of the pharmacologic activity of verapamil and accounts for 6% of excreted drug.

The steady-state plasma concentrations of norverapamil and verapamil are similar. Steady state after multiple once daily dosing is reached after three to four days.

Absorption: Greater than 90% of verapamil is rapidly absorbed from the small intestine after oral administration. Mean systemic availability of the unchanged compound after a single dose of non-retarded verapamil is 22% and that of retarded verapamil approximately 32%, owing to an extensive hepatic first-pass metabolism.

Bioavailability is about two times higher with repeated administration. Peak verapamil plasma levels are reached one to two hours after non-retarded administration, and four to five hours after retarded administration. The peak plasma concentration of norverapamil is attained approximately one and five hours after non-retarded or retarded administration, respectively.

The presence of food has no effect on the bioavailability of verapamil.

Distribution: Verapamil is widely distributed throughout the body tissues, the volume of distribution ranging from 1.8–6.8 l/kg in healthy subjects. Plasma protein binding of verapamil is approximately 90%.

Metabolism: Verapamil is extensively metabolised. *In vitro* metabolic studies indicate that verapamil is metabolised by

cytochrome P450 CYP3A4, CYP1A2, CYP2C8, CYP2C9 and CYP2C18. In healthy men, orally administered verapamil hydrochloride undergoes extensive metabolism in the liver, with 12 metabolites having been identified, most in only trace amounts. The major metabolites have been identified as various N and O-dealkylated products of verapamil. Of these metabolites, only norverapamil has any appreciable pharmacological effect (approximately 20% that of the parent compound), which was observed in a study with dogs.

Elimination: Following intravenous infusion, verapamil is eliminated bi-exponentially, with a rapid early distribution phase (half-life about four minutes) and a slower terminal elimination phase (half-life two to five hours).

Following oral administration, the elimination half-life is three to seven hours.

Approximately 50% of an administered dose is eliminated renally within 24 hours, 70% within five days. Up to 16% of a dose is excreted in the faeces. About 3% to 4% of renally excreted active substance is excreted unchanged. The total clearance of verapamil is nearly as high as the hepatic blood flow, approximately 1 l/h/kg (range: 0.7-1.3 l/h/kg).

There are large inter-individual differences in clearance.

Special populations

Paediatric patients

Limited information on the pharmacokinetics in the paediatric population is available. After intravenous dosing, the mean half-life of verapamil was 9.17 hours and the mean clearance was 30 l/h, whereas it is around 70 l/h for a 70-kg adult. Steady-state plasma concentrations appear to be somewhat lower in the paediatric population after oral dosing compared to those observed in adults.

Elderly patients

Aging may affect the pharmacokinetics of verapamil given to hypertensive patients. Elimination half-life may be prolonged in the elderly. The antihypertensive effect of verapamil was found not to be age-related.

Renal insufficiency

Impaired renal function has no effect on verapamil pharmacokinetics, as shown by comparative studies in patients with end-stage renal failure and subjects with healthy kidneys. Verapamil and norverapamil are not significantly removed by haemodialysis.

Hepatic insufficiency

The half-life of verapamil is prolonged in patients with impaired liver function owing to lower oral clearance and a higher volume of distribution.

EXCIPIENTS

Microcrystalline cellulose, Hypromellose, Lactose monohydrate, Macrogol 4000, Magnesium stearate, Sodium alginate, Povidone, Colloidal silicon dioxide, Ferric (III) hydroxide oxide, Titanium dioxide

Presentation: Blister pack of 10 x 10 tablets
Shelf life: Please refer to the outer box label.
Store in a cool, dry place below 30°C.
Keep out of reach of children.
Jauhi daripada kanak-kanak.

Manufactured by:

Salutas Pharma GmbH (a subsidiary of Hexal AG)
Barleben, Germany

For:

Hexal Pharma GmbH, Holzkirchen, Germany

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